Introduction to Robotics

Lecture 10: Dynamics of Open Chains

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Template made by Panagiotis Papagiannopoulos

Single Rigid Body Dynamics - Recap

- **Spatial Inertia Matrix:** $\mathcal{G}_b = \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & M\boldsymbol{I} \end{bmatrix} \in \mathbb{R}^{6\times 6}$
- **Spatial Momentum:** $\mathcal{P}_b = \mathcal{G}_b \mathcal{V}_b \in \mathbb{R}^6$
- **Adjoint operator:** $[ad_{\mathcal{V}}] = \begin{bmatrix} \hat{\boldsymbol{\omega}} & 0 \\ \hat{\boldsymbol{v}} & \hat{\boldsymbol{\omega}} \end{bmatrix}$, where $\mathcal{V} = \begin{bmatrix} \boldsymbol{\omega} \\ \boldsymbol{v} \end{bmatrix}$

Dynamics Equations:

$$\mathcal{F}_b = \mathcal{G}_b \dot{\mathcal{V}}_b - [ad_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b$$

Forward and Inverse Dynamics

- **Inverse Dynamics:** Given a desired acceleration $\dot{\mathcal{V}}_b$, we can compute the wrench to achieve it by

$$\mathcal{F}_b = \mathcal{G}_b \dot{\mathcal{V}}_b - [ad_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b$$

  - We can use this for control or trajectory synthesis
  - We answer the question: *"What is the wrench required to exert in order to achieve a desired acceleration?"*
- **Forward Dynamics:** Given a realized wrench $\mathcal{F}_b$, we can compute the acceleration generated by it

$$\dot{\mathcal{V}}_b = \mathcal{G}_b^{-1}\left(\mathcal{F}_b + [ad_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b\right)$$

  - We can use this for forward simulation or estimation
  - We answer the question: *"What is the acceleration that is generated by the application of a specific wrench?"*

Dynamics of Open Chains

$$\underbrace{\boldsymbol{M}(\boldsymbol{q})}_{\text{“Mass Matrix”}} \ddot{\boldsymbol{q}} + \underbrace{\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})}_{\text{“Coriolis/Gravity Forces”}} = \underbrace{\boldsymbol{\tau}}_{\text{“Joint torques/forces”}} + \underbrace{\boldsymbol{F}_{\text{ext}}}_{\text{“External forces”}}$$

- $\boldsymbol{q}$ are the joint angles
- we will ignore $\boldsymbol{F}_{\text{ext}}$ for now
- $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \underbrace{\boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}})}_{\text{“Coriolis forces”}} + \underbrace{\boldsymbol{g}(\boldsymbol{q})}_{\text{“Gravity forces”}}$
- This equation is also called the *“Manipulator Equation”*

Forward and Inverse Dynamics of Open Chains

- **Inverse Dynamics:** Given a desired acceleration $\ddot{\boldsymbol{q}}_d$, we can compute the wrench to achieve it by

$$\boldsymbol{\tau} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}_d + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \boldsymbol{F}_{\text{ext}}$$

  - We can use this for control or trajectory synthesis
  - We answer the question: *"What are the joint torques/forces required to exert in order to achieve some desired joint accelerations?"*
- **Forward Dynamics:** Given some realized joint torques/forces $\boldsymbol{\tau}$, we can compute the joint accelerations generated by them

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\Big(\boldsymbol{\tau} + \boldsymbol{F}_{\text{ext}} - \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\Big)$$

  - We can use this for forward simulation or estimation
  - We answer the question: *"What are the joint accelerations that are generated by the application of some specific joint torques/forces?"*

Lagrangian Derivation

We define the Lagrangian function as:

$$\mathcal{L}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \mathcal{K}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \mathcal{P}(\boldsymbol{q})$$

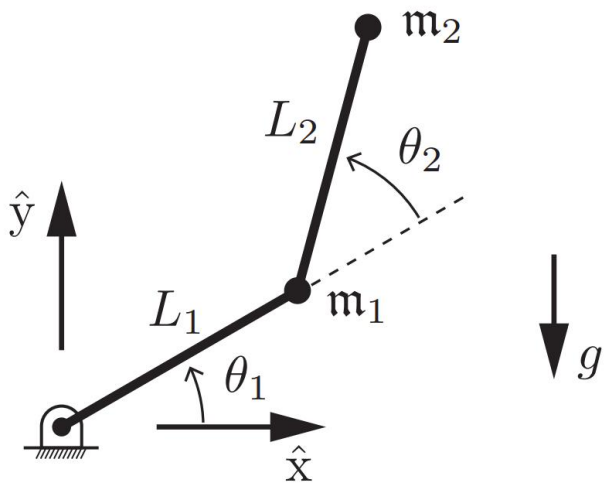
where

- $\mathcal{K}$ is the kinematic energy;
- $\mathcal{P}$ is the potential energy.

Now we can define the dynamics as:

$$\boldsymbol{\tau} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\boldsymbol{q}}} - \frac{\partial \mathcal{L}}{\partial \boldsymbol{q}}$$

Lagrangian Derivation - Example



Source: Modern Robotics: Mechanics, Planning, and Control, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press.

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 \\ L_1 \sin\theta_1 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

Lagrangian Derivation - Example (2)

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 \\ L_1 \sin\theta_1 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{bmatrix} = \begin{bmatrix} -L_1 \sin\theta_1 \\ L_1 \cos\theta_1 \end{bmatrix} \dot{\theta}_1$$

$$\begin{bmatrix} \dot{x}_2 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} -L_1 \sin\theta_1 - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

Lagrangian Derivation - Example (3)

We can now define the Lagrangian function as:

$$\mathcal{L} = \sum_{i=1}^{2} \left(\mathcal{K}_i - \mathcal{P}_i\right)$$

where

$$\mathcal{K}_1 = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2), \mathcal{P}_1 = m_1 g y_1$$
$$\mathcal{K}_2 = \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2), \mathcal{P}_2 = m_2 g y_2$$

And thus:

$$\boldsymbol{\tau}_1 = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} - \frac{\partial \mathcal{L}}{\partial \theta_1}$$
$$\boldsymbol{\tau}_2 = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_2} - \frac{\partial \mathcal{L}}{\partial \theta_2}$$

Lagrangian Derivation - Example (4)

After many trivial but tedius calculations, we can arrive to:

$$\boldsymbol{\tau} = \boldsymbol{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \underbrace{\boldsymbol{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \boldsymbol{g}(\boldsymbol{\theta})}_{\boldsymbol{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})}$$

where

$$\boldsymbol{M}(\boldsymbol{\theta}) = \begin{bmatrix} m_1 L_1^2 + m_2(L_1^2 + 2L_1 L_2 \cos\theta_2 + L_2^2) & m_2(L_1 L_2 \cos\theta_2 + L_2^2) \\ m_2(L_1 L_2 \cos\theta_2 + L_2^2) & m_2 L_2^2 \end{bmatrix}$$

$$\boldsymbol{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) = \begin{bmatrix} -m_2 L_1 L_2 \sin\theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ m2 L_1 L_2 \dot{\theta}_1^2 \sin\theta_2 \end{bmatrix}$$

$$\boldsymbol{g}(\boldsymbol{\theta}) = \begin{bmatrix} (m_1 + m_2) L_1 g \cos\theta_1 + m_2 g L_2 \cos(\theta_1 + \theta_2) \\ m2 g L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

Interesting Properties/Considerations

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{\tau}$$

- We moved from maximal coordinates (i.e. computing quantities for each body) to *reduced coordinates* (e.g. joint space)
  - Less variables
  - Joint constraints are enforced by construction
- $\boldsymbol{M}(\boldsymbol{q})$ is a symmetric positive-definite matrix
- $\dot{\boldsymbol{M}}(\boldsymbol{q}) - 2\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is a skew symmetric matrix
- $\mathcal{K} = \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}} \geqslant 0$

Newton-Euler Inverse Dynamics - Notation

- The Lagrangian approach is tractable only for simple systems

Newton-Euler Inverse Dynamics - Notation

- The Lagrangian approach is tractable only for simple systems
- We have one frame per body (at the CoM): $\{i\}$
- We denote the transformation of frame $\{i\}$ in frame $\{j\}$ at the *home* configuration as $\boldsymbol{M}_{i,j} \in SE(3)$
- $\boldsymbol{M}_i = \boldsymbol{M}_{i,0}$, $\boldsymbol{M}_{i-1,i} = \boldsymbol{M}_{i-1}^{-1}\boldsymbol{M}_i$ and $\boldsymbol{M}_{i,i-1} = \boldsymbol{M}_i^{-1}\boldsymbol{M}_{i-1}$
- Screw axis $\mathcal{S}_i$ expressed in frame $\{i\}$ is $\mathcal{A}_i = [Ad_{\boldsymbol{M}_i^{-1}}]\mathcal{S}_i$
- We denote the transformation of frame $\{i\}$ in frame $\{j\}$ for any configuration as $\boldsymbol{T}_{i,j}(\boldsymbol{q}) \in SE(3)$:
  - $\boldsymbol{T}_{i-1,i}(\boldsymbol{q}_i) = \boldsymbol{M}_{i-1,i}\exp(\mathcal{A}_i\boldsymbol{q}_i)$
  - $\boldsymbol{T}_{i,i-1}(\boldsymbol{q}_i) = \exp(-\mathcal{A}_i\boldsymbol{q}_i)\boldsymbol{M}_{i,i-1}$

Newton-Euler Inverse Dynamics - Notation (2)

- Twist of link $\{i\}$ in frame $\{i\}$: $\mathcal{V}_i = \begin{bmatrix} \omega_i & v_i \end{bmatrix}^T$
- Wrench transmitted through joint i to link frame $\{i\}$ expressed in frame $\{i\}$: $\mathcal{F}_i = \begin{bmatrix} \tau_i & f_i \end{bmatrix}^T$
- $\mathcal{G}_i \in \mathbb{R}^{6\times 6}$ spatial inertia of link $\{i\}$ expressed in frame $\{i\}$:

$$\mathcal{G}_i = \begin{bmatrix} \mathcal{I}_i & 0 \\ 0 & m_i \boldsymbol{I} \end{bmatrix}$$

where $\mathcal{I}_i \in \mathbb{R}^{3\times 3}$ is the rotational inertia matrix of link i and $\boldsymbol{I} \in \mathbb{R}^{3\times 3}$ is the identity matrix.

Newton-Euler Inverse Dynamics

- We can now recursively compute velocities and accelerations starting from the base frame and moving to the tip(s):

$$\mathcal{V}_i = \mathcal{A}_i \dot{\boldsymbol{q}}_i + [Ad_{\boldsymbol{T}_{i,i-1}}]\mathcal{V}_{i-1}$$
$$\dot{\mathcal{V}}_i = \mathcal{A}_i \ddot{\boldsymbol{q}}_i + [Ad_{\boldsymbol{T}_{i,i-1}}]\dot{\mathcal{V}}_{i-1} + [ad_{\mathcal{V}_i}]\mathcal{A}_i \dot{\boldsymbol{q}}_i$$

- Once we have computed all $\mathcal{V}_i$ and $\dot{\mathcal{V}}_i$, we need to find all the forces acting on each link:
  - We know from the single rigid body analysis that the total wrench given $\mathcal{V}_i, \dot{\mathcal{V}}_i$ is: $\mathcal{G}_i\dot{\mathcal{V}}_i - [ad_{\mathcal{V}_i}]^T\mathcal{G}_i\mathcal{V}_i$
  - The total wrench is the sum of the wrench transmitted through joint i to the link ($\mathcal{F}_i$) and the wrench transmitted through joint $i+1$ ($\mathcal{F}_{i+1}$)

$$\mathcal{G}_i\dot{\mathcal{V}}_i - [ad_{\mathcal{V}_i}]^T\mathcal{G}_i\mathcal{V}_i = \mathcal{F}_i - [Ad_{\boldsymbol{T}_{i+1,i}}]^T\mathcal{F}_{i+1}$$

- We now solve from the tip(s) to the base and get all $\mathcal{F}_i$
- Finally we have $\boldsymbol{\tau}_i = \mathcal{F}_i^T\mathcal{A}_i$

Newton-Euler Inverse Dynamics Algorithm

Base frame quantities:

$$\mathcal{V}_0 = \mathbf{0}$$
$$\dot{\mathcal{V}}_0 = \begin{bmatrix} \mathbf{0} & -\boldsymbol{g} \end{bmatrix}^T$$

where n the DoFs of the robot, and $\boldsymbol{g} \in \mathbb{R}^3$ is the gravity vector.

Forward Pass $(i = 1 \rightarrow n)$

$$\boldsymbol{T}_{i,i-1} = \exp(-\mathcal{A}_i \boldsymbol{q}_i)\boldsymbol{M}_{i,i-1}$$
$$\mathcal{V}_i = \mathcal{A}_i \dot{\boldsymbol{q}}_i + [Ad_{\boldsymbol{T}_{i,i-1}}]\mathcal{V}_{i-1}$$
$$\dot{\mathcal{V}}_i = \mathcal{A}_i \ddot{\boldsymbol{q}}_i + [Ad_{\boldsymbol{T}_{i,i-1}}]\dot{\mathcal{V}}_{i-1} + [ad_{\mathcal{V}_i}]\mathcal{A}_i \dot{\boldsymbol{q}}_i$$

Backward Pass $(i = n \rightarrow 1)$

$$\mathcal{F}_i = [Ad_{\boldsymbol{T}_{i+1,i}}]^T \mathcal{F}_{i+1} + \mathcal{G}_i \dot{\mathcal{V}}_i - [ad_{\mathcal{V}_i}]^T \mathcal{G}_i \mathcal{V}_i$$
$$\boldsymbol{\tau}_i = \mathcal{F}_i^T \mathcal{A}_i$$

where $\mathcal{F}_{n+1}$ is the wrench the end-effector applies to the environment.

Closed Form Derivation

- We want "closed-form" equations for $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}$, $\boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}})$ and $\boldsymbol{g}(\boldsymbol{q})$

$$\mathcal{V}_{\text{all}} = \begin{bmatrix} \mathcal{V}_1 & \cdots & \mathcal{V}_n \end{bmatrix}^T \in \mathbb{R}^{6n}$$

$$\mathcal{F}_{\text{all}} = \begin{bmatrix} \mathcal{F}_1 & \cdots & \mathcal{F}_n \end{bmatrix}^T \in \mathbb{R}^{6n}$$

$$\mathcal{A}_{\text{all}} = \begin{bmatrix} \mathcal{A}_1 & 0 & \cdots & 0 \\ 0 & \mathcal{A}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \mathcal{A}_n \end{bmatrix} \in \mathbb{R}^{6n\times 6n}$$

$$\mathcal{G}_{\text{all}} = \begin{bmatrix} \mathcal{G}_1 & 0 & \cdots & 0 \\ 0 & \mathcal{G}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \mathcal{G}_n \end{bmatrix} \in \mathbb{R}^{6n\times 6n}$$

Closed Form Derivation (2)

$$[ad_{\mathcal{V}_{\text{all}}}] = \begin{bmatrix} [ad_{\mathcal{V}_1}] & 0 & \cdots & 0 \\ 0 & [ad_{\mathcal{V}_2}] & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & [ad_{\mathcal{V}_n}] \end{bmatrix} \in \mathbb{R}^{6n \times 6n}$$

$$[ad_{\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}}}] = \begin{bmatrix} [ad_{\mathcal{A}_1 \boldsymbol{q}_1}] & 0 & \cdots & 0 \\ 0 & [ad_{\mathcal{A}_2 \boldsymbol{q}_2}] & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & [ad_{\mathcal{A}_n \boldsymbol{q}_n}] \end{bmatrix} \in \mathbb{R}^{6n \times 6n}$$

$$\mathcal{W}(\boldsymbol{q}) = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ [Ad_{\boldsymbol{T}_{2,1}}] & 0 & \cdots & 0 & 0 \\ 0 & [Ad_{\boldsymbol{T}_{3,2}}] & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & \cdots & \cdots & [Ad_{\boldsymbol{T}_{n,n-1}}] & 0 \end{bmatrix} \in \mathbb{R}^{6n \times 6n}$$

Closed Form Derivation (3)

$$\mathcal{W}(\boldsymbol{q}) = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ [Ad_{\boldsymbol{T}_{2,1}}] & 0 & \cdots & 0 & 0 \\ 0 & [Ad_{\boldsymbol{T}_{3,2}}] & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & \cdots & \cdots & [Ad_{\boldsymbol{T}_{n,n-1}}] & 0 \end{bmatrix} \in \mathbb{R}^{6n \times 6n}$$

$$\mathcal{L}(\boldsymbol{q}) = (\boldsymbol{I} - \mathcal{W}(\boldsymbol{q}))^{-1} = \begin{bmatrix} \boldsymbol{I} & 0 & 0 & \cdots & 0 \\ [Ad_{\boldsymbol{T}_{2,1}}] & \boldsymbol{I} & 0 & \cdots & 0 \\ [Ad_{\boldsymbol{T}_{3,1}}] & [Ad_{\boldsymbol{T}_{3,2}}] & \boldsymbol{I} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ [Ad_{\boldsymbol{T}_{n,1}}] & [Ad_{\boldsymbol{T}_{n,2}}] & [Ad_{\boldsymbol{T}_{n,3}}] & \cdots & \boldsymbol{I} \end{bmatrix} \in \mathbb{R}^{6n \times 6n}$$

Closed Form Derivation (4)

$$\mathcal{V}_{\text{base}} = \begin{bmatrix} [Ad_{\boldsymbol{T}_{1,0}}]\mathcal{V}_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{6n}$$

$$\dot{\mathcal{V}}_{\text{base}} = \begin{bmatrix} [Ad_{\boldsymbol{T}_{1,0}}]\dot{\mathcal{V}}_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{6n}$$

$$\mathcal{F}_{\text{tip}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ [Ad_{\boldsymbol{T}_{n+1,n}}]^T \mathcal{F}_{n+1} \end{bmatrix} \in \mathbb{R}^{6n}$$

Closed Form Derivation (5)

Now we can compute:

$$\begin{aligned}
\mathcal{V}_{\text{all}} &= \mathcal{L}(\boldsymbol{q})\Big(\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}} + \mathcal{V}_{\text{base}}\Big) \\
\dot{\mathcal{V}}_{\text{all}} &= \mathcal{L}(\boldsymbol{q})\Big(\mathcal{A}_{\text{all}}\ddot{\boldsymbol{q}} - [ad_{\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}}}]\mathcal{W}(\boldsymbol{q})\mathcal{V} - [ad_{\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}}}]\mathcal{V}_{\text{base}} + \dot{\mathcal{V}}_{\text{base}}\Big) \\
\mathcal{F}_{\text{all}} &= \mathcal{L}^T(\boldsymbol{q})\Big(\mathcal{G}_{\text{all}}\dot{\mathcal{V}}_{\text{all}} - [ad_{\mathcal{V}_{\text{all}}}]^T\mathcal{G}_{\text{all}}\mathcal{V}_{\text{all}} + \mathcal{F}_{\text{tip}}\Big) \\
\boldsymbol{\tau} &= \mathcal{A}_{\text{all}}^T\mathcal{F}_{\text{all}}
\end{aligned}$$

And:

$$\begin{aligned}
\boldsymbol{M}(\boldsymbol{q}) &= \mathcal{A}_{\text{all}}^T\mathcal{L}^T(\boldsymbol{q})\mathcal{G}_{\text{all}}\mathcal{L}(\boldsymbol{q})\mathcal{A}_{\text{all}} \\
\boldsymbol{c}(\boldsymbol{q},\dot{\boldsymbol{q}}) &= -\mathcal{A}_{\text{all}}^T\mathcal{L}^T(\boldsymbol{q})\Big(\mathcal{G}_{\text{all}}\mathcal{L}(\boldsymbol{q})[ad_{\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}}}]\mathcal{W}(\boldsymbol{q}) + [ad_{\mathcal{V}_{\text{all}}}]^T\mathcal{G}_{\text{all}}\Big)\mathcal{L}(\boldsymbol{q})\mathcal{A}_{\text{all}}\dot{\boldsymbol{q}} \\
\boldsymbol{g}(\boldsymbol{q}) &= \mathcal{A}_{\text{all}}^T\mathcal{L}^T(\boldsymbol{q})\mathcal{G}_{\text{all}}\mathcal{L}(\boldsymbol{q})\dot{\mathcal{V}}_{\text{base}}
\end{aligned}$$

Dynamics of Open Chains: Verdict

- The Lagrangian formulation is suitable only for simple systems
- The Newton-Euler formulation provides an efficient recursive algorithm
- The "closed-form" formulation is still a recursive algorithm
- There exist more sophisticated and more efficient algorithms (e.g. Roy Featherstone's Rigid Body Dynamics Algorithms book)
- We can even analytically compute the gradients of the dynamics equations

Bibliography

Chapter 8 from **Modern Robotics: Mechanics, Planning, and Control**, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press. ebook

Thank you

- **Any Questions?**
- **Office Hours**:
  - **Tue-Thu** *(10:00-12:00)*
  - 24/7 by email (`costashatz@upatras.gr`, subject: *ECE_IR_AM*)
- **Material and Announcements**

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